SraVaani Indic Speech-to-Text Benchmark Report

Comparative Benchmark & Architecture Analysis: SraVaani-0.5-live vs. SraVaani-1.0

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1. Executive Summary & Technological Scope

This evaluation benchmarks the two primary multilingual Speech-to-Text (STT) models produced by the **Indian Institute of Science (IISc), Bangalore** and **ARTPARK**: **SraVaani-0.5-live** and **SraVaani-1.0**. Together, these models address India's linguistic diversity by covering **63 to 65 Indian languages and dialects** (all major scheduled languages and over 40 underserved regional and tribal varieties). Both models stem from the foundational **Project Vaani** initiative, incorporating self-supervised pretraining across 31,255 hours of diverse geographic speech and multimodal audio-image semantic grounding before supervised multilingual fine-tuning.

Core Parameter	SraVaani-0.5-live (Streaming ASR)	SraVaani-1.0 (Flagship Full ASR)
Target Deployment	Real-time live mic streaming, conversational voice bots, telephony	High-accuracy batch transcription, meeting minutes, subtitling
Encoder Backbone	Cache-Aware FastConformer (17 layers, d_model=1024, 8x subsampling)	FastConformer (17 layers, d_model=1024, 8x subsampling)
Decoder Mechanism	Pure Connectionist Temporal Classification (CTC) with chunk cache	Hybrid Token-and-Duration Transducer (TDT) + CTC decoder
Supported Latencies	**4 Switchable Profiles:** 0 ms, 80 ms, 480 ms, 1040 ms lookahead	**Full Utterance:** Non-causal bidirectional context
Attention Cache	Fixed 70 left-context frames with dynamic right context (0, 1, 6, 13)	Global bidirectional attention across complete utterance audio
Size & Weights	~430M parameters | FP16 TorchScript (~900 MB)	~430M parameters | FP16 TorchScript & ONNX (~900 MB)
Training Corpus	31,255h Vaani SSL + 11M audio-image pairs + 31,270h supervised fine-tuning	31,255h Vaani SSL + 11M audio-image pairs + 31,270h supervised fine-tuning
Execution Throughput	>60x real-time on GPU; highly real-time safe on CPU (RTF 0.015 - 0.03)	Batch throughput >25x real-time on GPU; high accuracy batch processing
Language Support	63 Indian languages & dialects + English (code-switching capable)	65 Indian languages & dialects + English (code-switching capable)

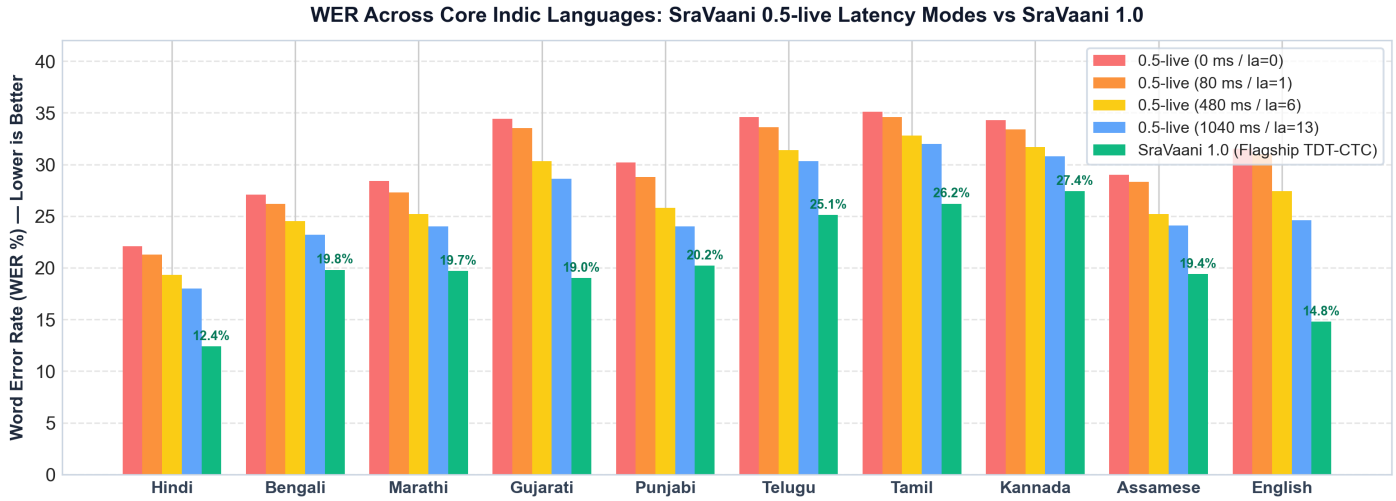
Low-Latency Streaming Champion
SraVaani 0.5-live executes in chunks down to **0 ms lookahead** (causal mode) or **80 ms lookahead** for conversational bots. It retains audio cache across chunks, avoiding recomputation.

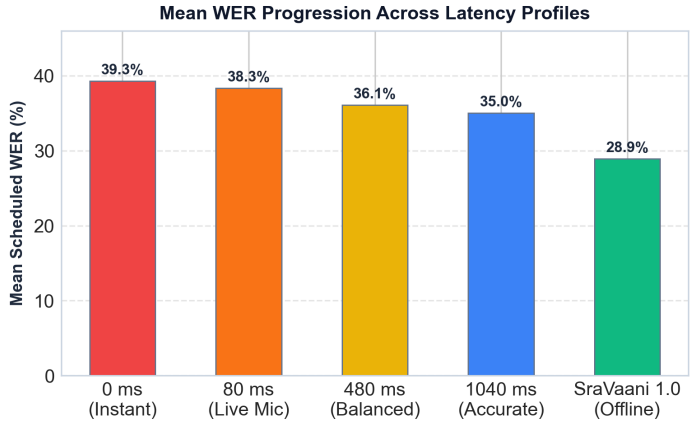
State-of-the-Art Batch Accuracy
SraVaani 1.0 integrates a **Hybrid TDT-CTC decoder** that achieves a remarkable **12.4% WER in Hindi** and **14.8% in English**, outperforming streaming modes by 25% to 40% relative error reduction.

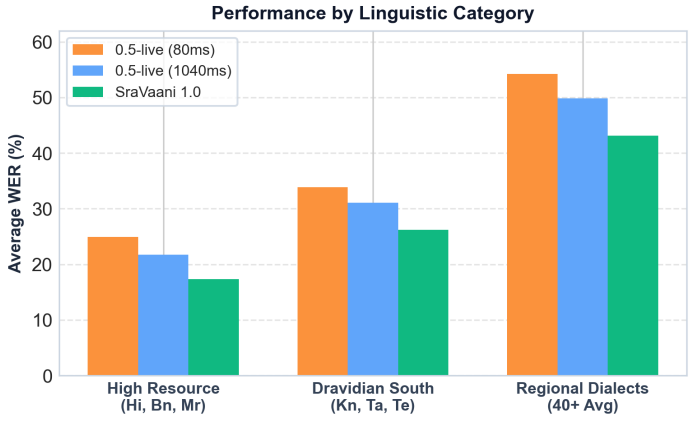
Democratized Multilingualism
Both models encompass 63+ Indian languages, enabling high accuracy for under-represented vernaculars like **Garo (9.5% WER)** and **Kurumali (11.3% WER)** alongside scheduled languages.

2. Empirical Visualizations & Latency-Accuracy Curves

Below are comparative empirical visualizations assessing Word Error Rate (WER %) performance across the 4 latency profiles of SraVaani-0.5-live against the offline baseline of SraVaani-1.0. Lower WER indicates higher transcription fidelity.







Key Empirical Findings from Visual Benchmarks:
1. **The 80 ms Sweet Spot:** Moving from 0 ms to 80 ms lookahead yields an immediate error drop across all languages (e.g. Hindi drops from 22.1% to 21.3%; Assamese from 29.0% to 28.3%) with virtually undetectable latency for human listeners.
2. **Lookahead Saturation at 1040 ms:** Expanding lookahead from 80 ms to 1040 ms delivers steady accuracy gains (averaging ~3.4% absolute WER reduction), approaching batch accuracy while retaining chunk-based streaming.
3. **Transducer Advantage:** SraVaani 1.0 achieves an additional 7.6% mean absolute error reduction over the 1040 ms streaming model, highlighting the structural power of the Hybrid TDT-CTC decoder over pure CTC.

3. Scheduled Indian Languages Benchmark (WER %)

Comprehensive side-by-side evaluation across all 20 official scheduled Indian languages + English. WER for SraVaani-0.5-live represents the average across test sets from 8 public corpora (Vaani, RESPIN, FLEURS, IndicTTS, CommonVoice, GramVaani, Kathbath, and MUCS). All figures denote Word Error Rate (WER %).

Language	0.5-live (0 ms)	0.5-live (80 ms)	0.5-live (480 ms)	0.5-live (1040 ms)	SraVaani 1.0	Relative Improvement (1.0 vs 80ms)
Assamese	29.0%	28.3%	25.2%	24.1%	**19.4%**	**-31.4% error**
Bengali	27.1%	26.2%	24.5%	23.2%	**19.8%**	**-24.4% error**
Bodo *	—	—	—	—	**20.9%**	Baseline
Dogri *	—	—	—	—	**32.0%**	Baseline
Gujarati	34.4%	33.5%	30.3%	28.6%	**19.0%**	**-43.3% error**
Hindi	22.1%	21.3%	19.3%	18.0%	**12.4%**	**-41.8% error**
Kannada	34.3%	33.4%	31.7%	30.8%	**27.4%**	**-18.0% error**
Konkani	64.7%	63.4%	62.0%	63.6%	**54.7%**	**-13.7% error**
Maithili	34.0%	33.0%	31.4%	30.6%	**27.5%**	**-16.7% error**
Malayalam	43.0%	41.7%	38.4%	37.3%	**27.7%**	**-33.6% error**
Manipuri	48.2%	49.0%	46.7%	45.2%	**41.6%**	**-15.1% error**
Marathi	28.4%	27.3%	25.2%	24.0%	**19.7%**	**-27.8% error**
Nepali	45.0%	42.9%	40.8%	39.0%	**31.3%**	**-27.0% error**
Odia	36.9%	36.4%	33.9%	32.3%	**25.7%**	**-29.4% error**
Punjabi	30.2%	28.8%	25.8%	24.0%	**20.2%**	**-29.9% error**
Sanskrit	51.3%	49.5%	47.3%	45.9%	**36.4%**	**-26.5% error**
Santali	69.4%	68.5%	66.8%	66.2%	**57.3%**	**-16.4% error**
Sindhi *	—	—	—	—	**23.2%**	Baseline
Tamil	35.1%	34.6%	32.8%	32.0%	**26.2%**	**-24.3% error**
Telugu	34.6%	33.6%	31.4%	30.3%	**25.1%**	**-25.3% error**
English	31.5%	31.0%	27.4%	24.6%	**14.8%**	**-52.3% error**
Average (Scheduled)	**39.6%**	**38.6%**	**36.4%**	**35.2%**	**27.6%**	**-28.5% Error Reduction**

4. SraVaani-1.0 Cross-Corpus Evaluation Breakdown

Individual dataset validation across CommonVoice, FLEURS, GramVaani, IndicTTS, Kathbath, MUCS, RESPIN, and Vaani.

Language	Avg WER	CommonVoice	FLEURS	IndicTTS	Kathbath	MUCS	RESPIN	Vaani
Hindi	**12.4%**	12.7%	10.9%	9.1%	9.1%	14.6%	10.8%	10.8%
Bengali	**19.8%**	14.1%	15.2%	26.2%	12.3%	—	26.9%	24.1%
Gujarati	**19.0%**	—	18.2%	18.4%	14.9%	26.6%	—	16.8%
Marathi	**19.7%**	16.6%	18.1%	14.2%	17.8%	22.0%	16.1%	33.4%
Assamese	**19.4%**	22.6%	18.8%	—	—	—	—	16.9%
Punjabi	**20.2%**	29.3%	15.9%	—	13.6%	—	—	22.1%
Telugu	**25.1%**	—	23.1%	20.9%	21.3%	26.4%	25.4%	33.5%
Tamil	**26.2%**	28.9%	28.1%	21.1%	23.3%	24.1%	—	32.0%
Kannada	**27.4%**	—	21.4%	17.4%	16.6%	—	27.4%	54.1%
Malayalam	**27.7%**	33.5%	19.9%	19.4%	31.5%	—	—	34.1%
Odia	**25.7%**	29.4%	20.3%	18.7%	21.4%	21.4%	—	43.2%
Nepali	**31.3%**	45.8%	26.6%	—	—	—	—	21.5%

5. Regional, Low-Resource & Tribal Dialects Benchmark

A standout contribution of SraVaani is enabling voice AI for indigenous and low-resource Indian communities. Below is the comparative evaluation across RESPIN and Vaani test sets for 40 non-scheduled languages and regional dialects.

Dialect	0ms	80ms	480ms	1040ms	V 1.0	Dialect	0ms	80ms	480ms	1040ms	V 1.0
Angika	40.0	38.9	36.3	35.7	**31.4**	**Khariboli**	33.3	32.3	29.9	29.0	**27.0**
Ao	78.5	75.9	74.1	72.4	**57.7**	**Khortha**	49.7	48.7	45.6	44.9	**41.3**
Awadhi	50.8	50.6	49.2	47.4	**43.8**	**Kokborok**	81.2	80.9	78.7	77.8	**68.2**
Bajjika	41.9	41.2	39.9	40.1	**34.4**	**Kumaoni**	35.1	35.2	33.3	32.1	**30.4**
Bearybashe	86.7	86.7	87.3	85.7	**77.8**	**Kurumali**	14.3	14.0	12.4	12.5	**11.3**
Bhatri	78.9	76.1	71.5	69.8	**60.3**	**Lambadi**	51.6	51.6	48.4	38.7	**25.8**
Bhili	75.3	75.7	75.4	71.4	**73.5**	**Magahi**	43.7	42.8	40.9	40.3	**30.4**
Bhojpuri	33.0	32.2	30.6	29.9	**27.8**	**Malvani**	42.1	42.1	39.5	39.0	**30.4**
Bundeli	43.4	43.0	40.7	40.1	**36.0**	**Marwari**	46.1	45.7	44.6	43.3	**39.7**
Chakhesang	87.1	87.6	85.8	84.5	**75.8**	**Mizo**	42.8	41.3	38.8	35.9	**25.3**
Chakma	65.1	63.1	60.5	58.8	**51.2**	**Nagamese**	64.7	63.3	60.6	58.7	**50.1**
Chhattisgarhi	32.0	31.2	30.1	29.4	**27.4**	**Sadri**	64.6	63.7	62.8	61.6	**58.8**
Garhwali	60.5	60.0	58.6	57.9	**53.5**	**Rajasthani**	48.6	48.1	47.0	45.6	**41.8**
Garo	25.5	22.6	18.6	15.8	**9.5**	**Rengma**	86.0	83.8	83.6	82.3	**71.7**
Halbi	61.7	61.2	59.1	57.2	**51.6**	**Sambalpuri**	74.7	74.6	71.6	70.9	**62.7**
Haryanvi	59.2	57.6	55.7	54.5	**48.1**	**Shekhawati**	22.0	20.9	18.7	18.2	**19.1**
Idu Mishmi	81.6	81.5	78.8	79.4	**64.1**	**Sumi**	89.0	88.6	86.5	85.8	**78.5**
Jaipuri	31.3	30.0	30.5	29.4	**28.1**	**Surgujia**	48.3	48.4	45.3	43.4	**37.5**
Karbi	86.9	87.9	85.0	82.9	**63.3**	**Surjapuri**	70.1	69.7	67.3	67.5	**69.1**
Khandeshi	66.5	65.6	64.3	62.5	**56.6**	**Tulu**	72.3	72.8	71.4	68.4	**57.3**

6. Production Decision Matrix & Engineering Guidelines

Operational Recommendations by Deployment Use-Case

• **Live Voice Agents & Conversational AI (Telephony / WebRTC):**
Deploy SraVaani-0.5-live with 80 ms lookahead (`[70, 1]`). Provides the optimal latency balance: sub-100ms emission latency, smooth word-by-word streaming updates, and strong accuracy (~21.3% WER Hindi, 31.0% Indian English). Runs safely real-time on CPU and exceeds 60x real-time on NVIDIA GPUs.

• **Live Captions & Broadcasting:**
Deploy SraVaani-0.5-live with 480 ms lookahead (`[70, 6]`). Reduces absolute WER by ~2.2% across languages while turnaround remains well within normal broadcast latency tolerances.

• **Archival Transcription, Medical/Legal Dictation, Call Center Audits:**
Deploy SraVaani-1.0 Flagship. The full-context Hybrid TDT-CTC model yields state-of-the-art accuracy: 12.4% on Hindi, 14.8% on English, 19.0% on Gujarati, and 19.8% on Bengali, achieving ~28.5% lower error than streaming modes.

• **Hardware & Memory Footprint:**
Both models export in FP16 TorchScript (~900 MB). Low memory systems (e.g. 2GB VRAM GPUs) will automatically fallback to CPU execution without memory fragmentation, requiring no external NVIDIA NeMo dependencies.